

Customer Intelligence: Segmentation using K-Means & PCA

Advanced Marketing Analytics & Behavioral Clustering

Python

Scikit-Learn

PCA

K-Means

Feature Engineering

Project Overview

This project focuses on identifying distinct customer segments within a retail dataset to drive data-driven marketing strategies. By leveraging **Unsupervised Machine Learning**, specifically K-Means clustering enhanced by Principal Component Analysis (PCA), we transform raw transactional data into actionable customer personas.

Key Features & Engineering

Unlike basic clustering models, this project implements robust feature engineering to capture true customer value:

- **Customer Tenure:** Calculated active days from `Dt_Customer`.
- **Total Spending:** Aggregated revenue across Wines, Fruits, Meat, Fish, Sweets, and Gold products.
- **Household Dynamics:** Derived total children from combined `Kidhome` and `Teenhome` data.
- **Data Validation:** Automated cleaning of unrealistic age and income outliers to ensure model reliability.

Technical Workflow

1. **Exploratory Data Analysis (EDA):** Visualizing distributions and handling missing values.
2. **Preprocessing:** Standardizing features using `StandardScaler` to ensure uniform feature influence.

3. **Optimal Cluster Selection:** Using the **Elbow Method** and **Silhouette Analysis** to determine the ideal number of segments ($k=4$).
4. **Dimensionality Reduction:** Applying PCA to visualize high-dimensional clusters in a 2D space.
5. **Cluster Profiling:** Generating heatmaps to understand the unique characteristics of each segment (e.g., High-Spenders vs. Deal-Seekers).

Visualizations

- **Elbow Plot:** To identify the "knee" for optimal K.
- **PCA Scatter Plot:** Visual representation of segment separation.
- **Cluster Heatmap:** A comprehensive look at average metrics per cluster.